European Semiconductor Financial Benchmarking

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Data Science / Statistical Research

Research question

How do six European semiconductor companies compare in growth, profitability, innovation spending and balance-sheet strength?

Data and methods

Financial-statement panel for ASML, ASM International, Besi, AIXTRON, VAT Group and Technoprobe. Main sample: 2021–2025, 30 company-years, with 2020 as an auxiliary base year.

Six financial ratios, annual median benchmarks, IQR screening and PCA on five standardized indicators; original workbook and source records retained.

Selected results

- Median revenue growth: 31.70% (2021), -5.33% (2023) and 11.07% (2025).
- PCA explains 46.17% on PC1 and 26.47% on PC2 (72.65% combined).
- IQR screening highlights Besi 2021 and VAT Group 2022, without removing them.

Interpretation and limitations

Six firms do not constitute a representative industry sample. Different reporting policies and currencies matter, and the PCA describes correlations rather than causes.

Reproduce

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Rscript scripts/run_all.R
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Repository and full documentation